ALCO EXECUTIVE REPORT

Aurelia Bank Treasury & ALM
Risk Control Tower

IRRBB, liquidity, FX and hedge decision support for a fictional banking book

AS OF	REPORTING UNIT	DATA POSTURE
19 AUG 2026	**TRY MILLION**	**CONTROLLED SYNTHETIC**

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Executive decision

The analysis points to one immediate formal remediation and one near-term liquidity protection action.

33.3%	78.1%	13.7%
FX OPEN / EQUITY	**RAPID-RUN LCR PROXY**	**WORST EVE LOSS**
20.0% formal limit	7-day survival | exploratory	15.0% internal limit

Decision requested

- Approve an 80% FX hedge target using an FX swap or forward, subject to independent pricing, counterparty, collateral and post-trade limit controls.
- Preserve HQLA and pre-position contingent funding because combined-stress LCR proxy headroom is exhausted and the survival horizon reaches the 30-day floor.
- Activate the rapid digital-run contingency workstream: rehearse a seven-day response, protect unencumbered HQLA and monitor intraday deposit-outflow triggers.
- Authorize validation of indicative TRY pay-fixed IRS sizing before any execution decision; the model targets a 50% DV01 reduction, not a pre-approved trade.

Control distinction

The FX position is the only formal engine limit breach. Combined and rapid-run liquidity results are management stress indicators; the formal control uses the base proxy.

Decision logic: remediate the formal FX breach, activate rapid-run readiness, preserve the liquidity buffer and validate the TRY duration hedge before execution.

Data architecture and methodology

Official evidence, internal assumptions and controlled synthetic bank records are deliberately segregated.

PUBLIC EVIDENCE	CONTROLLED BANKING BOOK	DECISION OUTPUTS
TCMB policy rate and FX BDDK sector benchmark Basel IRRBB shock calibration	149 positions 1,156 contractual cash flows Synthetic curves and behavioral assumptions	EVE and NII sensitivity Liquidity stress and survival FX exposure and hedge overlay
Official observation / methodology	Deterministic synthetic	ALCO decision support

IRRBB

Economic value sensitivity is calculated under all six Basel-prescribed scenarios using the revised currency shock calibration effective from 1 January 2026. The adverse loss is divided by Tier 1 capital. Twelve-month NII sensitivity uses parallel up and down shocks and governed behavioral assumptions.

Liquidity

HQLA, 30-day net cash outflows, available stable funding and required stable funding are combined into transparent LCR and NSFR proxies. The outputs are educational portfolio analytics and are not represented as regulatory returns.

Market context

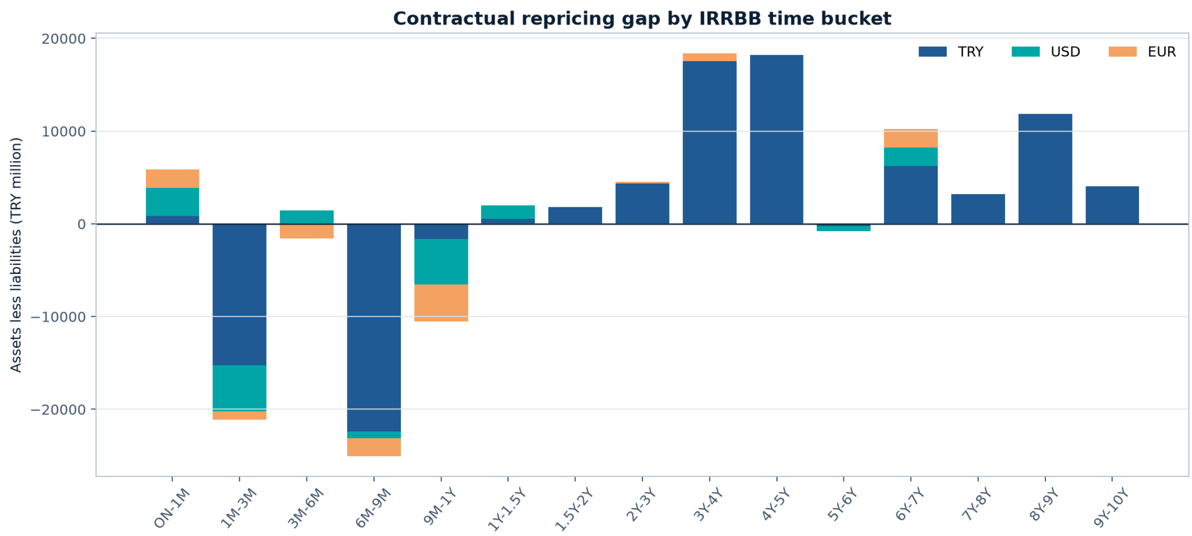
The data cutoff uses a 37.0% TCMB policy rate, USD/TRY buying rate of 47.8467 and EUR/TRY buying rate of 55.5061 on 19 August 2026. The BDDK June 2026 snapshot is used only as sector context.

Basel Committee - Recalibration of IRRBB shocks (d578)
TCMB - Policy rate announcement
TCMB via e-Devlet - Official exchange rates
BDDK - Monthly banking sector bulletin

Balance sheet and repricing structure

A deposit-funded TRY 180bn balance sheet carries negative repricing gaps inside the first year.

Metric	TRY mn	Interpretation
Total assets	180,000	TRY 120bn loans, TRY 35bn securities and TRY 25bn cash/interbank
Total liabilities	157,500	Deposits are 76.5% of liabilities
Tier 1 capital	22,500	Basis for EVE and FX appetite ratios
Accounting identity	0.0	Assets minus liabilities minus equity

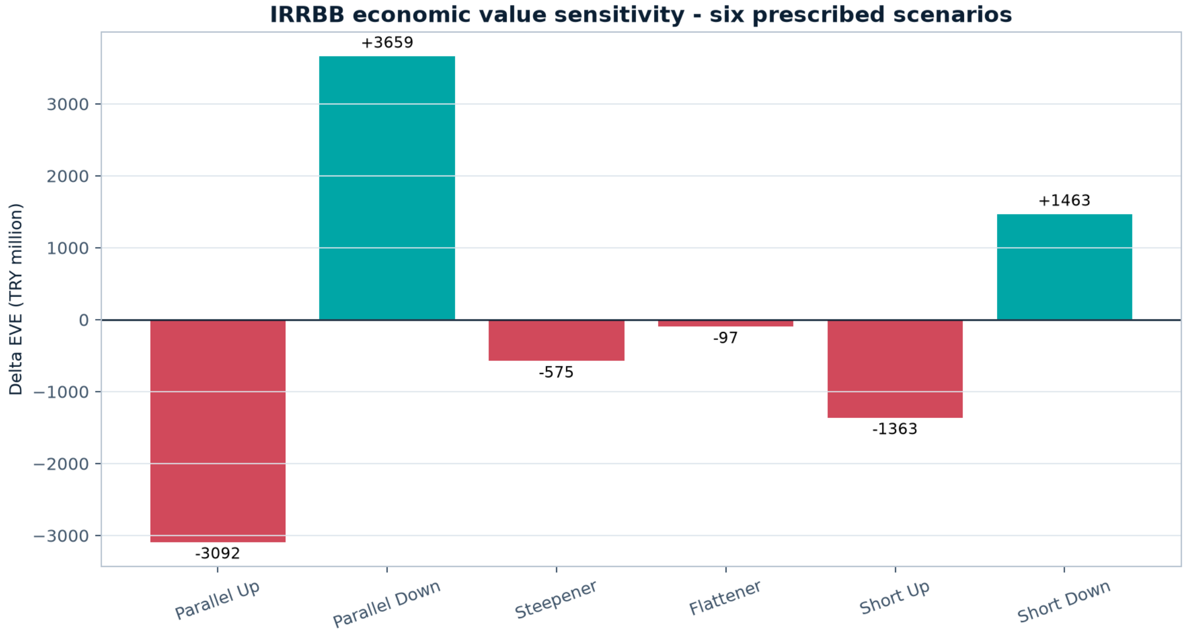


Interpretation

TRY liabilities reprice faster than assets across several sub-one-year buckets. The pattern explains why parallel-down is the adverse NII case even though long-dated assets later restore cumulative sensitivity.

IRRBB: economic value sensitivity

Parallel-up is the binding EVE shock, with modest remaining appetite headroom.



TRY 37.39bn	-TRY 3.09bn	13.7%
BASELINE EVE	**WORST DELTA EVE**	**LOSS / TIER 1**
All currencies	Parallel-up	15.0% limit

ALCO implication

The position remains within the 15.0% limit, but only about 1.3 percentage points of headroom remain. The largest sensitivity is in TRY; indicative pay-fixed IRS sizing is therefore a second-stage action after data and price validation.

[Basel Committee - IRRBB shock calibration]

IRRBB: 12-month NII sensitivity

NII stays within appetite, but behavioral assumptions remain decision-critical.

Scenario	Baseline NII	Shocked NII	Delta	Delta %
Parallel up	22,791.4	24,116.4	1,324.9	5.8%
Parallel down	22,791.4	21,466.5	(1,324.9)	(5.8%)

WHAT DRIVES THE RESULT

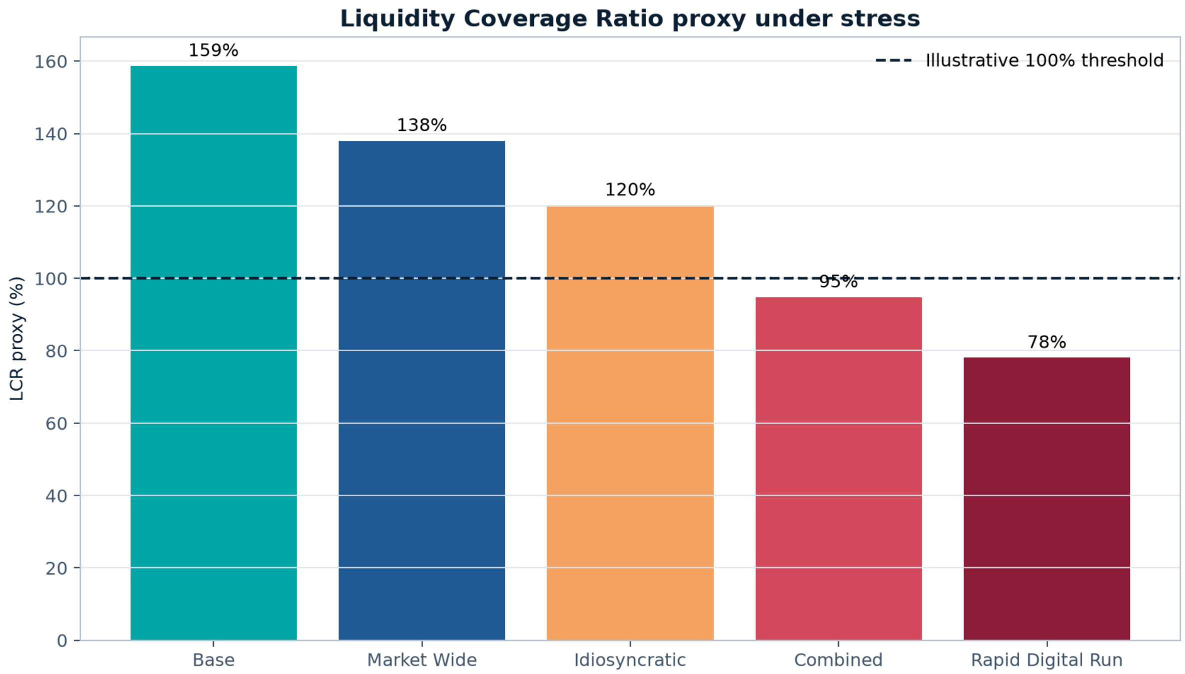
Deposit beta, repricing lag, rate floors and customer migration determine how fast non-maturity deposits respond to a rate shock.

ALCO implication

The 5.8% absolute sensitivity is comfortably below the 12.0% appetite limit. Hedge volume should not be increased solely on this result; behavioral assumptions must first be back-tested and independently challenged.

Liquidity stress and structural funding

A rapid digital deposit run compresses the survival horizon to seven days while structural funding remains strong.



158.7%	78.1%	7 days
BASE LCR PROXY	**RAPID-RUN LCR PROXY**	**RAPID-RUN SURVIVAL**
Formal control within limit	Severe exploratory stress	Contingency horizon

Management action

The combined proxy remains 94.8% with a 30-day survival floor. Under the severe rapid-run scenario, 85% of 30-day outflows arrive by day 7 and HQLA loses TRY 1.46bn of market value. Retain the buffer, pre-position collateral and funding, rehearse the contingency plan and establish intraday deposit-outflow triggers. The NSFR proxy remains 143.4%, confirming that the principal pressure is short-term stress rather than structural funding.

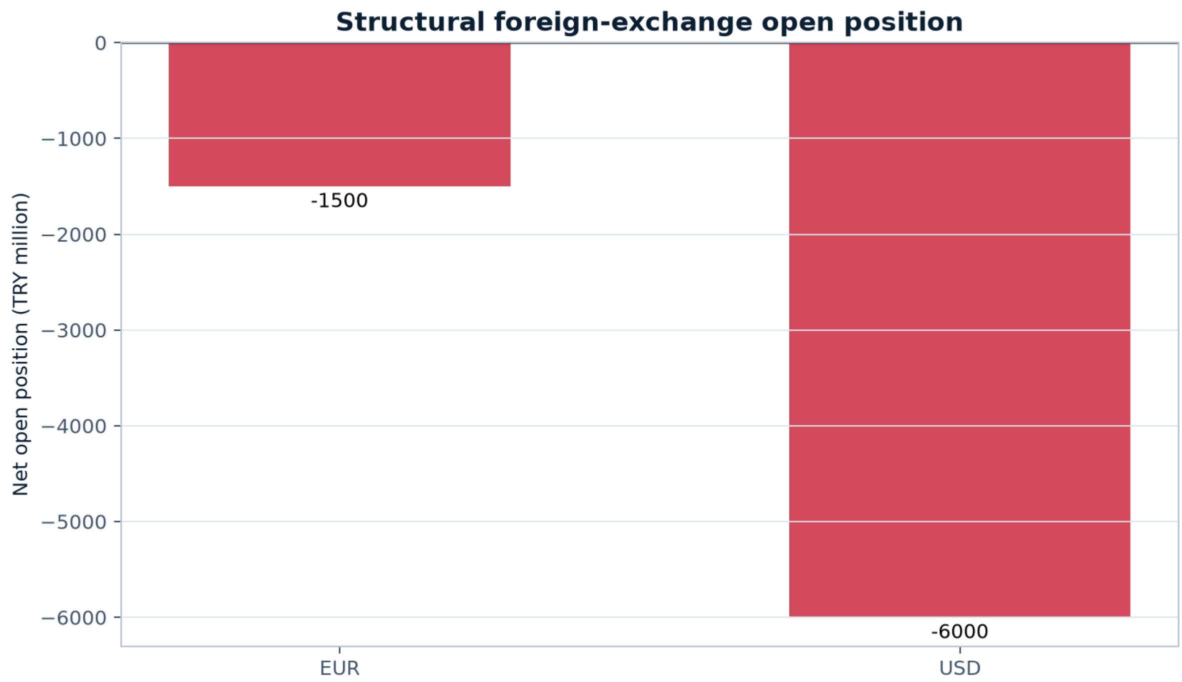
Regulatory boundary

These calculations are transparent portfolio proxies built from synthetic records. The rapid-run case is an exploratory management stress, not a Basel-prescribed scenario. None of the outputs include every jurisdiction-specific reporting rule or may be used as regulatory submissions.

[BIS - Liquidity Coverage Ratio framework]
[BIS - Net Stable Funding Ratio framework]

FX exposure and hedge recommendations

The only formal breach is directly remediable, subject to ALCO and execution controls.



Risk	CCY	Instrument	Direction	Notional TRY mn	Target
IRRBB_DV01	EUR	plain vanilla interest rate swap	PAY FIXED RECEIVE FLOAT	438.5	50% reduction
IRRBB_DV01	TRY	plain vanilla interest rate swap	PAY FIXED RECEIVE FLOAT	10,316.6	50% reduction
IRRBB_DV01	USD	plain vanilla interest rate swap	RECEIVE FIXED PAY FLOAT	163.5	50% reduction
FX_OPEN_POSITION	EUR	fx swap or forward	BUY FX SELL TRY	1,200.0	80% reduction
FX_OPEN_POSITION	USD	fx swap or forward	BUY FX SELL TRY	4,800.0	80% reduction

Execution gate

ALCO approval -> independent price validation -> counterparty, legal and collateral checks -> trade capture -> post-trade EVE, NII, liquidity and FX revalidation. Every model recommendation is marked ALCO_REVIEW_REQUIRED.

Controls, engineering quality and governance

The control tower distinguishes model integrity from business risk and preserves a reproducible audit trail.

ID	Control	Status	Observed	Limit
RL01	six prescribed EVE scenarios are present	WITHIN_LIMIT	6.000000	6.000000
RL02	maximum EVE decline / Tier 1	WITHIN_LIMIT	13.743894	15.000000
RL03	absolute delta NII	WITHIN_LIMIT	5.813365	12.000000
RL04	base LCR proxy	WITHIN_LIMIT	158.721676	100.000000
RL05	NSFR proxy	WITHIN_LIMIT	143.353978	100.000000
RL06	combined survival horizon	WITHIN_LIMIT	30.000000	30.000000
RL07	aggregate FX open position / equity	BREACH	33.333333	20.000000

10 / 10
DATA QUALITY
All controls pass

46
UNIT TESTS
Pytest suite

93.88%
LINE COVERAGE
Above 90% gate

Engineering guardrails

- Deterministic data seed and explicit data classifications
- Configuration-driven assumptions, limits and shock parameters
- Checksum manifest, artifact verification and read-only API
- Ruff, Pytest, coverage threshold, GitHub Actions and CodeQL

Control principle: a red risk indicator is evidence that the control tower is working - not a reason to paint the dashboard green.

ALCO actions, limitations and source register

A controlled sequence turns analysis into accountable action while preserving the model's governance boundaries.

0-5 DAYS	5-15 DAYS	15-30 DAYS
Hedge FX Approve 80% target Set execution limits Confirm counterparties	**Activate rapid-run plan** Retain HQLA Pre-position funding Rehearse seven-day response	**Validate TRY IRS** Challenge DV01 mapping Independent pricing Approve or resize

Limitations

- Aurelia Bank is fictional and its positions, curves and cash flows are deterministic synthetic records.
- LCR and NSFR outputs are transparent proxies and omit jurisdiction-specific reporting adjustments.
- The hedge layer is a risk-reduction heuristic, not an optimizer, pricing engine, accounting treatment or execution system.
- Non-maturity deposit betas, prepayment and runoff assumptions require empirical calibration before production use.
- Rapid-run timing, runoff and HQLA haircut assumptions are severe exploratory overlays that require independent validation and reverse-stress calibration.

Primary source register

Basel Committee - Recalibration of IRRBB shocks
BIS - Liquidity Coverage Ratio framework
BIS - Net Stable Funding Ratio framework
TCMB - Electronic Data Delivery System
TCMB - Policy rate announcement
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DECISION	Approve the remediation sequence, assign Treasury, ALM, Market Risk and Finance owners, and require a documented post-trade rerun.